

Block-encoding K matrices A_k of fixed size $256 \times 256 = 2^8 \times 2^8$

$K = k^3$ for k in $[2, 3, 4, 5, 6]$ (so K in $[8, 27, 64, 125, 216]$); phase_bitsize $b = 32$.

Approach 1 (SVD): $A_k = U_k \Sigma_k V_k$.

Cost = $2 * \text{BlockUnitaryInterferometerSynthesisQROAM}(K, 2^n)$
+ $\text{QROAMClean}((K, 2^n), b) + \text{ctrl-AddIntoPhaseGrad} + \text{QROAMCleanAdjoint}$.

Approach 2 (direct dilation): A_k completed to W_k of size $2^{(n+1)} \times 2^{(n+1)}$.

Cost = $\text{BlockUnitaryInterferometerSynthesisQROAM}(K, 2^{(n+1)})$. $\alpha = 1$.

Approach 3 (reflection rectangular): $B = H_A \cdot \text{SWAP} \cdot O_A \cdot H_A$ where

$O_A = \text{QROAMClean}((K, N, N), b) + \text{ctrl-AddIntoPhaseGrad} + \text{QROAMCleanAdjoint}$.

Subnormalization $\alpha = N = 256$ (algorithm depth grows by N over $\alpha=1$ schemes).

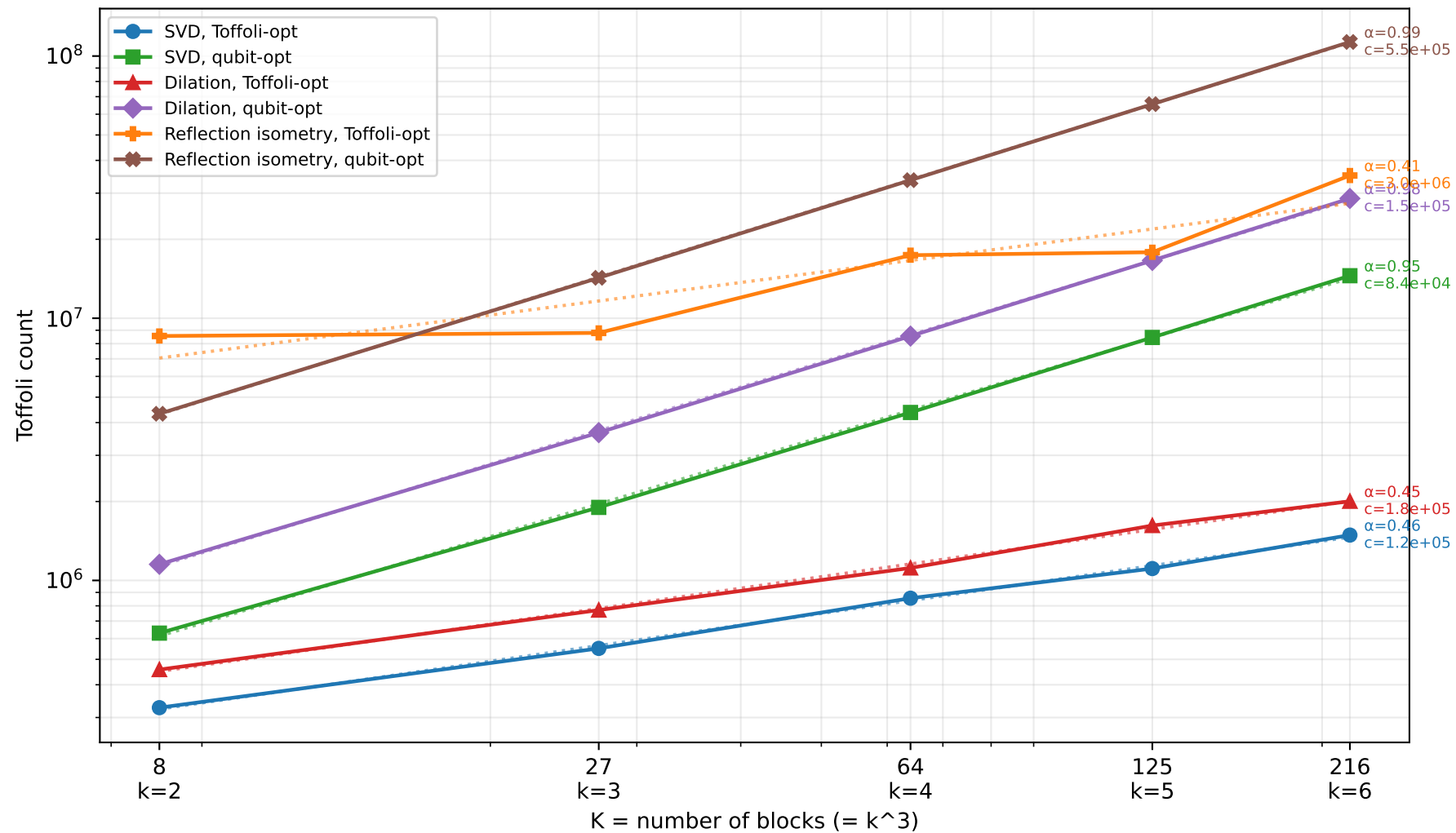
Each cost is optimized over its `log_block_size` sweep.

Two optimization modes per approach: Toffoli-optimal and qubit-optimal.

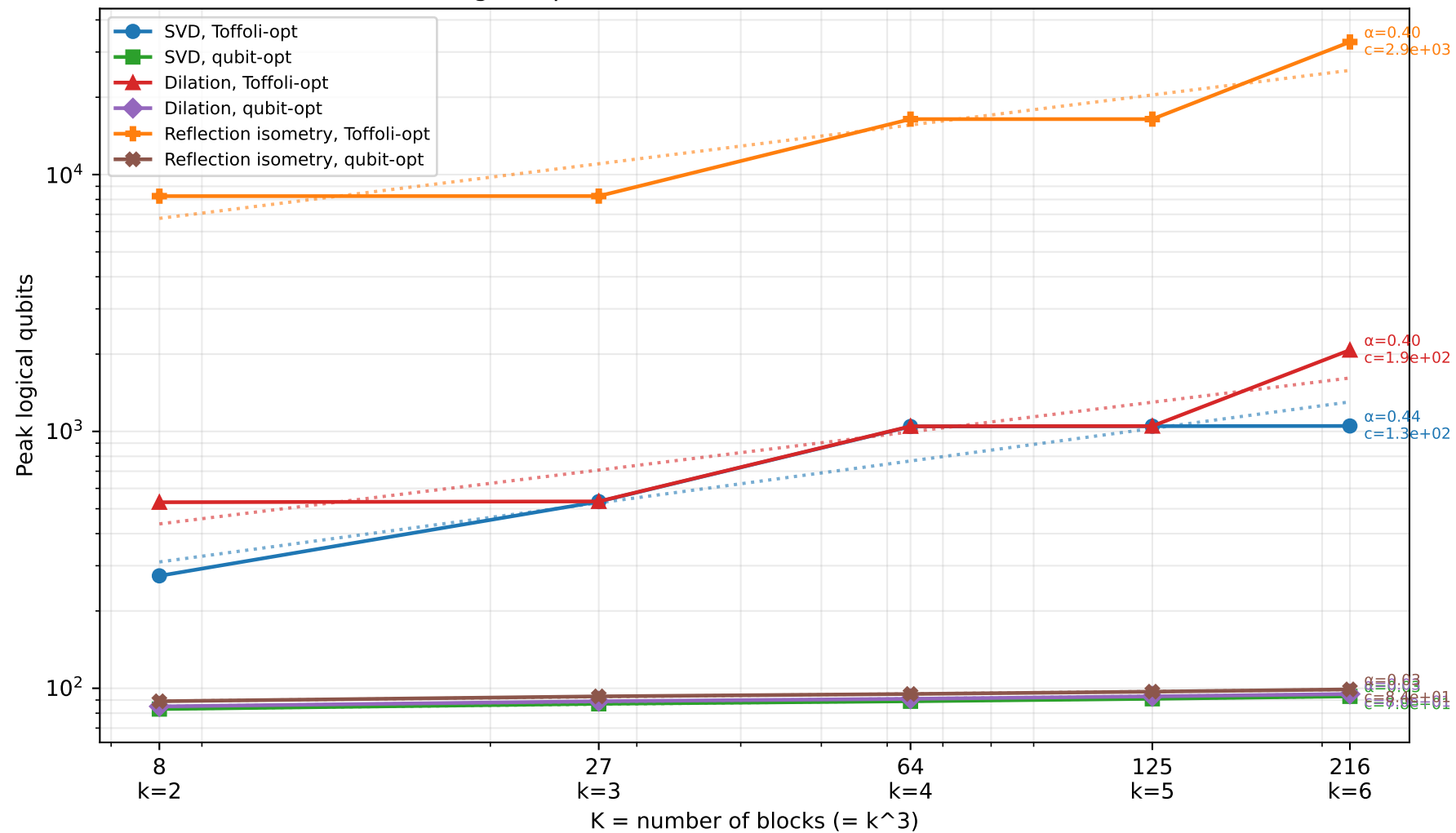
Fits $y = c * K^\alpha$:

SVD, Toffoli-opt	T: $\alpha=0.458$, $c=1.246e+05$; Q: $\alpha=0.435$, $c=1.255e+02$
SVD, qubit-opt	T: $\alpha=0.953$, $c=8.448e+04$; Q: $\alpha=0.034$, $c=7.754e+01$
Dilation, Toffoli-opt	T: $\alpha=0.454$, $c=1.751e+05$; Q: $\alpha=0.397$, $c=1.912e+02$
Dilation, qubit-opt	T: $\alpha=0.975$, $c=1.495e+05$; Q: $\alpha=0.033$, $c=7.952e+01$
Reflection isometry, Toffoli-opt	T: $\alpha=0.412$, $c=2.991e+06$; Q: $\alpha=0.403$, $c=2.923e+03$
Reflection isometry, qubit-opt	T: $\alpha=0.991$, $c=5.477e+05$; Q: $\alpha=0.031$, $c=8.350e+01$

Toffoli count: SVD vs direct dilation (N=256, b=32)



Peak logical qubits: SVD vs direct dilation (N=256, b=32)



Resource counts (fixed N=256, b=32)

k	K	SVD T-opt	SVD T-opt	SVD Q-opt	SVD Q-opt	Dil T-opt	Dil T-opt	Dil Q-opt	Dil Q-opt	Ref1 T-opt	Ref1 T-opt	Ref1 Q-opt	Ref1 Q-opt	alpha
2	8	327,112	274	629,538	83	456,990	530	1,152,055	85	8,556,032	8241	4,321,792	89	256
3	27	550,476	533	1,899,118	87	770,718	534	3,661,879	89	8,790,528	8245	14,263,80	93	256
4	64	855,424	1045	4,371,394	89	1,115,686	1047	8,549,431	91	17,399,29	164383	33,624,57	95	256
5	125	1,109,794	1048	8,447,292	91	1,619,302	1049	16,607,28	93	17,854,46	164406	65,543,68	97	256
6	216	1,489,176	1050	14,527,64	93	2,003,766	2074	28,628,02	95	35,023,36	328251	113,160,70	99	256